would once again find himself sitting in front of a console at MOX-2—though this time as ISRO chairman—as the countdown began for the final descent of the Chandrayaan-3 lander. Next to him, Nilesh Desai, director of the Space Applications Centre (SAC) in Ahmedabad, was conscious of the recent failure of Russia's Luna 25, which too was to land near the South Pole and worried about the "unknown unknowns", as Sivan called them. Somanath himself was unusually calm. Having seen 50 rocket launches since he joined ISRO in 1985, the sexagenarian says, "I do not have the habit of getting excited. We had done all that we could to ensure success. There was nothing more we could do and I remained confident." That supreme confidence—his own as well as the team's—came from the hard work they had put in to correct the flaws of the previous launch and prepare for any exigency, including the unknown unknowns.

This, then, is the inspirational story of how Team ISRO pulled itself from the depths of despair, against all odds, to script lunar history. Together, they would make India only the fourth country to land a spacecraft on the moon and the first to do so near the unchartered and treacherous South Pole.

THE DEPTH OF DESPAIR

Four years ago, when ISRO lost contact with Chandrayaan-2 during its descent to the lunar surface, it would take days to locate the fallen lander. Only after a photograph taken by its lunar orbiter, which, unlike the lander, happened to be doing its job perfectly, did scientists get to know that it had crash-landed and lay sprawled on the lunar surface. As per protocol, Sivan, as ISRO chairman, formed a national failure analysis committee (FAC), headed by the experienced and accomplished V. Narayanan, director of the Liquid Propulsion Systems Centre (LPSC) in Thiruvananthapuram. Narayanan's team had put together the propulsion system for Chandrayaan-2, and its failure had left him stunned and disappointed. He was given a month to look into the entire performance of Chandrayaan-2 and its systems, pinpoint the reasons for the setback and make recommendations for a successful landing the next time. At the end of the month, the team came out with a 900-page report on what went wrong and how to overcome those problems.

Unforgiving as space is, a deviation of even one per cent or a faulty system can end in catastrophic failure. That is what the committee discovered when it examined the reasons for Chandrayaan-2's failure. Sivan, who was still chairman, recalls, "There were issues with all the three major systems critical to the spacecraft—propulsion, navigation and guidance control."

SPACE'S ELEVEN

The key individuals behind one of the most spectacular space missions undertaken by India



S. SOMANATH, 60

Chairman, ISRO

ISRO's 'Captain Cool', Somanath joined the Vikram Sarabhai Space Centre in 1985 and later contributed to the development of the PSLV and the giant GSLV Mark III. Before becoming ISRO chairman in 2022, he had overseen the propulsion systems for the Chandrayaan-2 lander. Somanath studied mechanical engineering at Kollam's TKM College and has a Master's degree in aerospace engineering from IISc.



P. VEERAMUTHUVEL, 46

Project Director, Chandrayaan-3

Born in Tamil Nadu's Villupuram district, Veeramuthuvel has been part of some of India's toughest space missions, including the Mars Orbiter. He did an engineering degree from Chennai, then an M.Tech., and after a brief stint with HAL, joined ISRO in 2004 as a project engineer. Simultaneously, he completed his Ph.D. from IIT Madras. As project director for Chandrayaan-3, he oversaw coordination between key ISRO centres.



M. SANKARAN, 59

Director, UR Rao Satellite Centre, Bengaluru

As director of the lead centre for the design and development of ISRO satellites, Sankaran played a key role in developing the communication and power systems for the Chandrayaan and Mars Orbiter missions. He joined the URSC after obtaining a Master's degree in Physics in 1986 from the Bharathidasan University, Tiruchirappalli. He is now spearheading the avionics design, electronics and power systems of the Gaganyaan programme.